

Panasonic

Operating Instructions (Overall)

AC Servo Motor & Driver MINAS A5-series

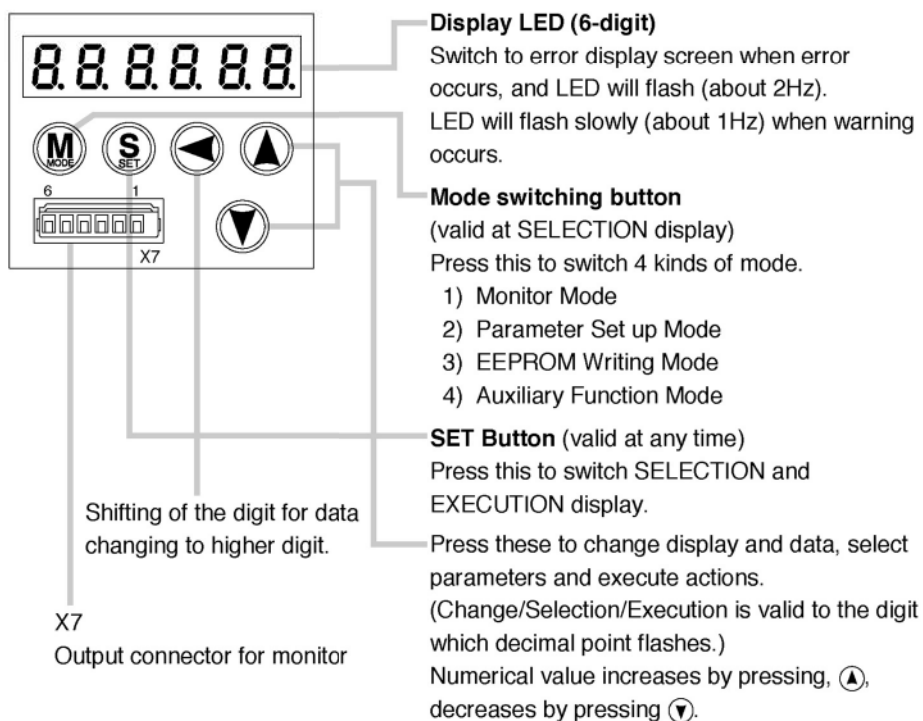


* This product image is 200W type of A5-series.

How to Use the Front Panel

Setup

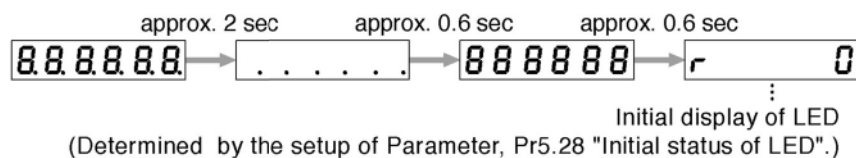
Setup with the Front Panel



Initial Status of the Front Panel Display (7 Segment LED)

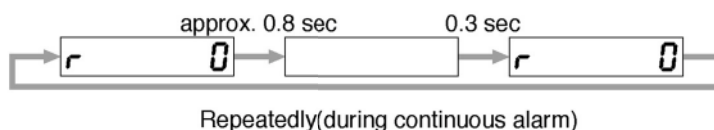
Status

Front panel display shows the following after turning on the power of the driver.



Upon Occurrence of an Alarm

If a driver alarm is generated, the front panel display shows the following repeatedly.



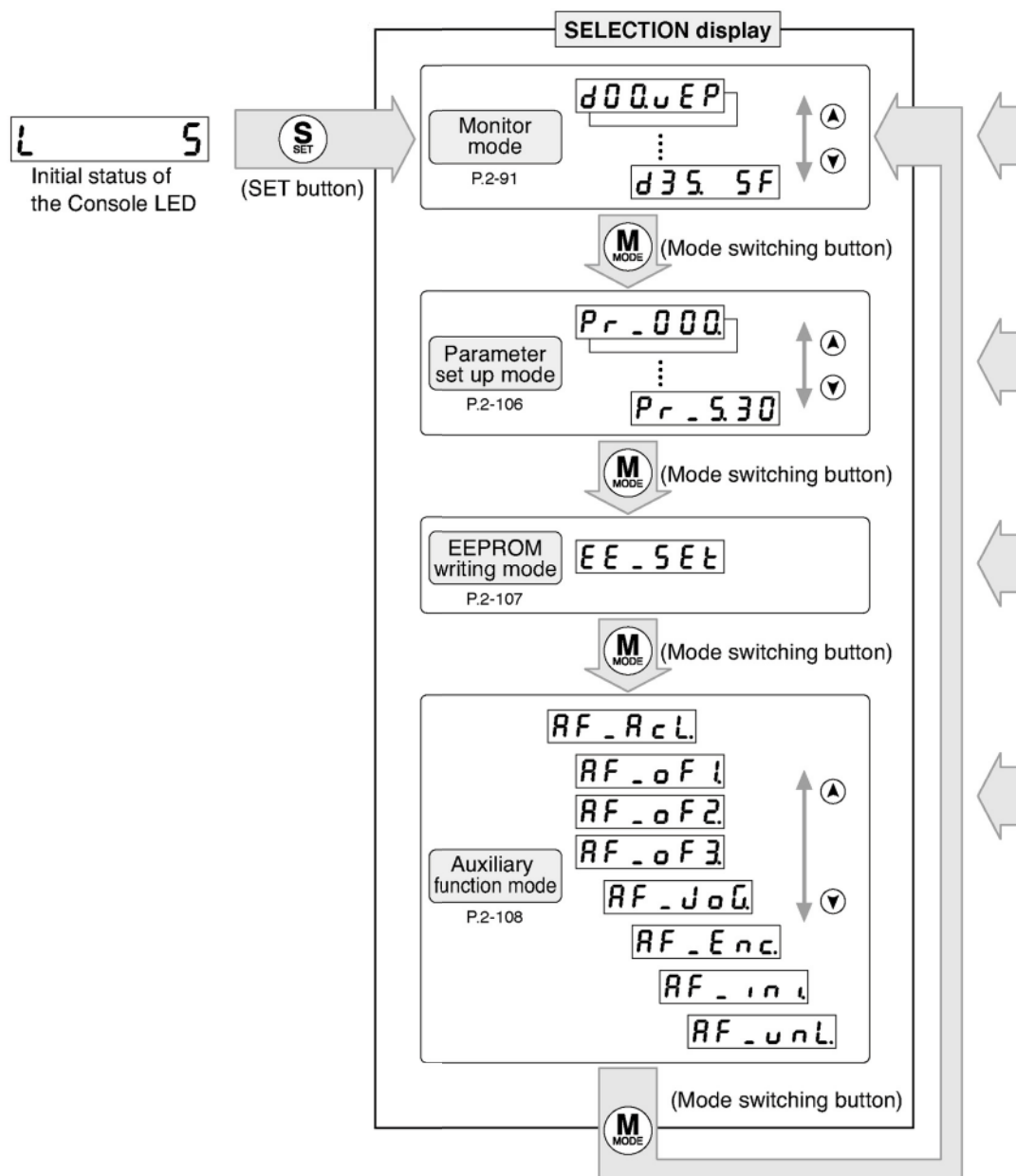
Below shows possible cause of an alarm.

alarm No.	Alarm	Content
A0	Overload protection	Load factor is 85% or more the protection level.
A1	Over-regeneration alarm	Regenerative load factor is 85% or more the protection level.
A2	Battery alarm	Battery voltage is 3.2 V or lower.
A3	Fan alarm	Fan has stopped for 1 sec.
A4	Encoder communication alarm	The number of successive encoder communication errors exceeds the specified value.
A5	Encoder overheat alarm	The encoder detects overheat alarm.
A6	Oscillation detection alarm	Oscillation or vibration is detected.
A7	Lifetime detection alarm	The life expectancy of capacity or fan becomes shorter than the specified time.
A8	External scale error alarm	The external scale detects the alarm.
A9	External scale communication alarm	The number of successive external scale communication errors exceeds the specified value.

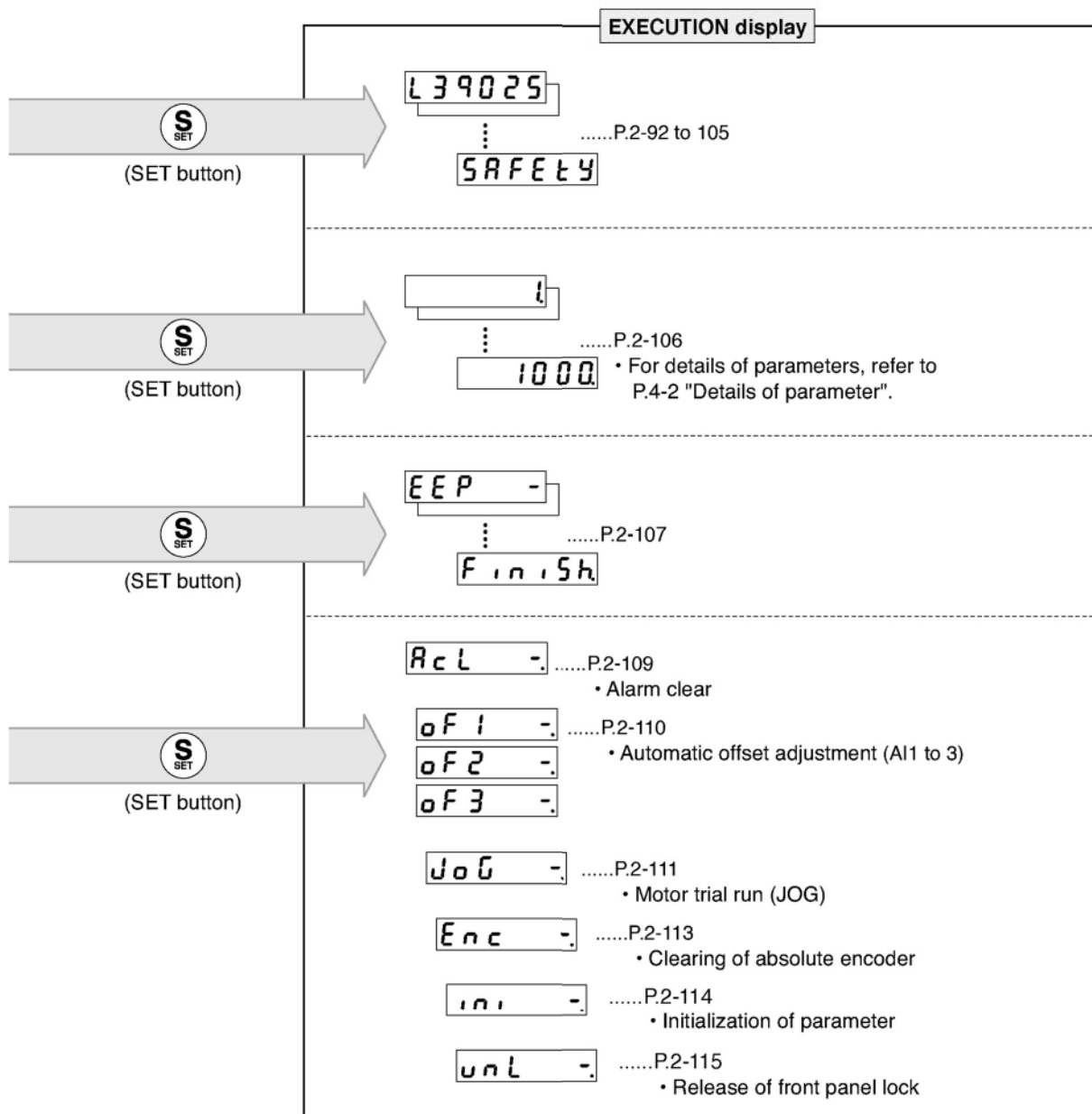
How to Use the Front Panel

Structure of Each Mode

Use each button on the touch panel to select the structure and switch the mode.



- Note** You can change the flashing decimal point with , then shift the digit for data change “.”
- Caution** On power-up, the monitor mode executed is displayed according to the setup of Pr5.28 LED initial status.



How to Use the Front Panel

Setup of front panel lock

Outline

To prevent operational error e.g. unintentional parameter modification, the front panel may be locked.

Once locked, operations on the panel are limited as follows:

Mode	Locked panel conditions
Monitor Mode	No limitation: all monitored data can be checked.
Parameter Set up Mode	No parameter can be changed but setting can be checked.
EEPROM Writing Mode	Cannot be run. (No display)
Auxiliary Function Mode	Cannot be run except for "Release of front panel lock". (No display)

How to operate

• Related parameters

Parameter No.		Title	Function
Class	No.		
5	35	Setup of front panel lock	Locks the operation attempted from the front panel.

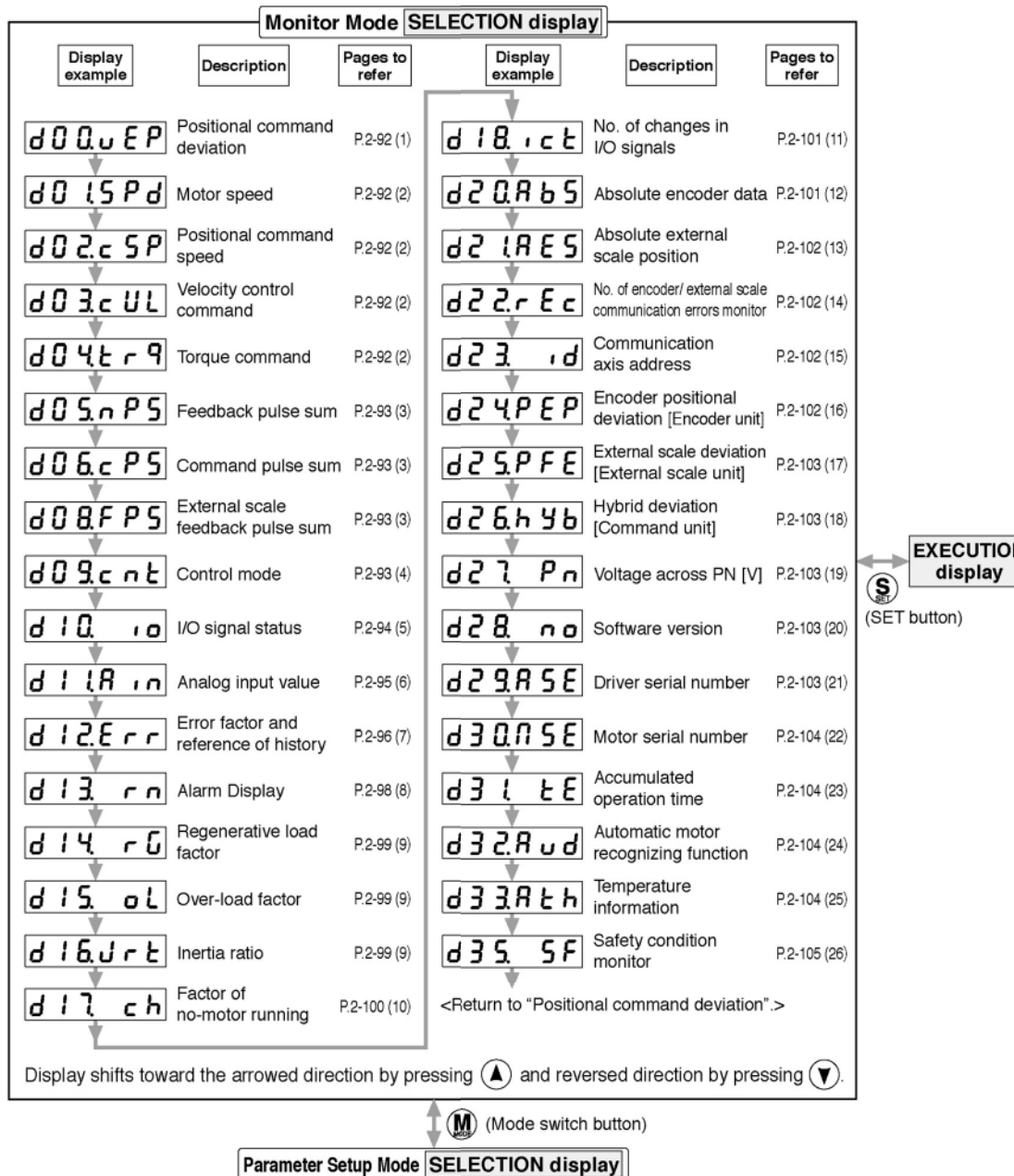
Lock and unlock can be made in one of two ways.

Procedure	Front panel	Setup support software PANATERM
Lock	(1) Set Pr5.35 "Front panel lock" to 1, and writ the setting to EEPROM. (2) Turn on power to the driver. (3) The front panel is locked.	
Unlock	(1) Execute the auxiliary function mode, front panel lock release function. (2) Turn on power to the driver. (3) The front panel is unlocked.	(1) Set Pr5.35 "Front panel lock" to 0, and writ the setting to EEPROM. (2) Turn on power to the driver. (3) The front panel is unlocked.

How to Use the Front Panel

Monitor Mode (SELECTION display)

To change the monitor display setting, select the display option to be changed from "SELECTION display", and press **(S)** to change to "EXECUTION display". After completion of changing, press **(S)** to return to the selection display,



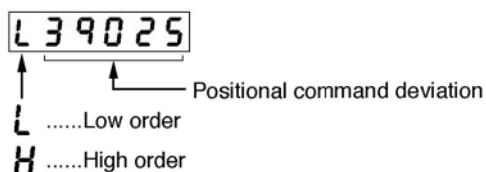
Note When you turn on the Product for the first time, display shows **r** (at motor stall). To change this display, change the setup of Pr5.28 (Initial status of LED).

How to Use the Front Panel

Monitor Mode (EXECUTION display)

(1) Display of positional command deviation [command unit]

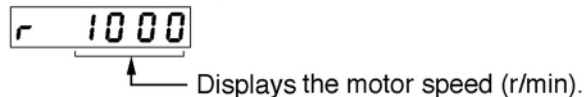
Displays positional deviation of the command unit in High order or Low order.



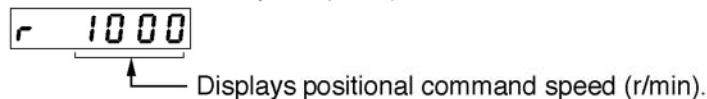
• To switch between Low order (L) and High order (H), press ◀.

(2) Display of motor speed, positional command speed, velocity control command and torque command

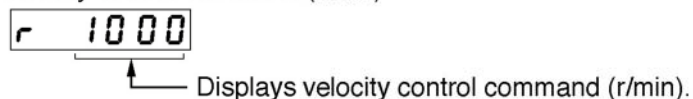
• Motor speed (r/min)



• Positional command speed (r/min)



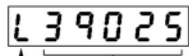
• Velocity control command (r/min)



• Torque command (%)



(3) Display of Feedback Pulse Sum, Command Pulse Sum and External Scale Feedback Pulse Sum**• Feedback Pulse Sum [Encoder feedback pulse]**



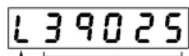
 LLow order

 HHigh order

• To switch between Low order (L) and High order (H), press .



• Command Pulse Sum [Command Pulse]



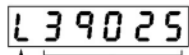
 LLow order

 HHigh order

• To switch between Low order (L) and High order (H), press .



• External Scale Feedback Pulse Sum



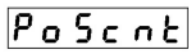
 LLow order

 HHigh order

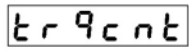
• To switch between Low order (L) and High order (H), press .

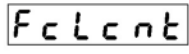


(4) Display of Control Mode

Position control mode

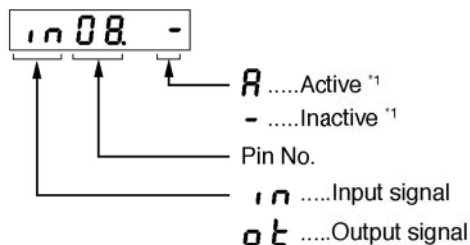
Velocity control mode

Torque control mode

Full-closed control mode

(5) Display of I/O Signal Status

Displays the control input and output signal to be connected to connector X4.
Use this function to check if the wiring is correct or not.

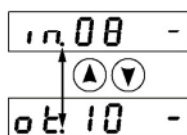


- Shift the flashing decimal point with ◀.

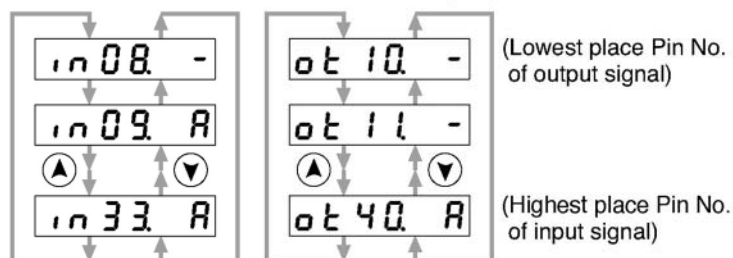
in 08 - (Right side of decimal point : Pin No. selection)

in 08 - (Left side of decimal point : Input/Output Pin No. selection)

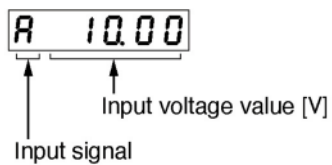
- Select In or Out by pressing ▲ or ▼ button.



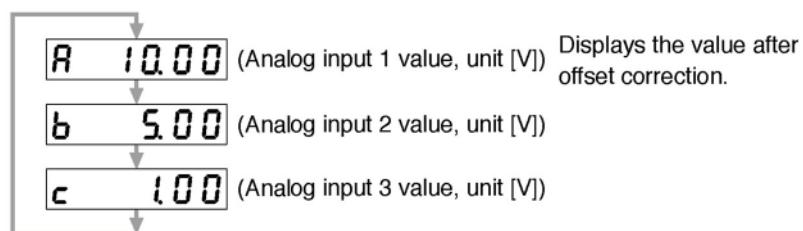
- Select the Pin No. to be monitored by pressing ▲ ▼.



- *1 When input signal Active : Input signal photocoupler is ON.
 Inactive : Input signal photocoupler is OFF.
 When output signal Active : Output signal transistor is ON.
 Inactive : Output signal transistor is OFF.

(6) Display of Analog Input Value

•Select the signal No. to be monitored by pressings .



Caution ⚠ Voltage exceeding $\pm 10\text{V}$ can not be displayed correctly.

(7) Display of Error Factor and Reference of History

Error code No. (- - - appears if no error occurs)

Err.Present error

E - 0History 0 (latest error)

E 13History 13 (oldest error)

• You can refer the last 14 error factors (including present one)
Press to select the factor to be referred.

<List of error code No.>

Error code		Protective function	Attribute		
Main	Sub		History	Can be cleared	Immediate stop
11	0	Control power supply under- voltage protection		☐	
12	0	Over-voltage protection	☐	☐	
13	0	Main power supply under-voltage protection (between P to N)		☐	
	1	Main power supply under-voltage protection (AC interception detection)		☐	
14	0	Over-current protection	☐		
	1	IPM error protection	☐		
15	0	Over-heat protection	☐		☐
16	0	Over-load protection	☐	☐ *1	
18	0	Over-regeneration load protection	☐		☐
	1	Over-regeneration Tr error protection	☐		
21	0	Encoder communication disconnect error protection	☐		
	1	Encoder communication error protection	☐		
23	0	Encoder communication data error protection	☐		
24	0	Position deviation excess protection	☐	☐	☐
	1	Velocity deviation excess protection	☐	☐	☐
25	0	Hybrid deviation excess error protection	☐		☐
26	0	Over-speed protection	☐	☐	☐
	1	2nd over-speed protection	☐	☐	
27	0	Command pulse input frequency error protection	☐	☐	☐
	2	Command pulse multiplier error protection	☐	☐	☐
28	0	Limit of pulse replay error protection	☐	☐	☐
29	0	Deviation counter overflow protection	☐	☐	
30	0	Safety detection		☐	
33	0	IF overlaps allocation error 1 protection	☐		
	1	IF overlaps allocation error 2 protection	☐		
	2	IF input function number error 1 protection	☐		
	3	IF input function number error 2 protection	☐		
	4	IF output function number error 1 protection	☐		
	5	IF output function number error 2 protection	☐		
	6	CL fitting error protection	☐		
	7	INH fitting error protection	☐		

Error code		Protective function	Attribute		
Main	Sub		History	Can be cleared	Immediate stop
34	0	Software limit protection	○	○	
36	0 to 2	EEPROM parameter error protection			
37	0 to 2	EEPROM check code error protection			
38	0	Over-travel inhibit input protection		○	
39	0	Analog input1 excess protection	○	○	○
	1	Analog input2 excess protection	○	○	○
	2	Analog input3 excess protection	○	○	○
40	0	Absolute system down error protection	○	○	
41	0	Absolute counter over error protection	○		
42	0	Absolute over-speed error protection	○	○	
43	0	Initialization failure	○		
44	0	Absolute single turn counter error protection	○		
45	0	Absolute multi-turn counter error protection	○		
47	0	Absolute status error protection	○		
48	0	Encoder Z-phase error protection	○		
49	0	Encoder CS signal error protection	○		
50	0	External scale connection error protection	○		
	1	External scale communication error protection	○		
51	0	External scale status 0 error protection	○		
	1	External scale status 1 error protection	○		
	2	External scale status 2 error protection	○		
	3	External scale status 3 error protection	○		
	4	External scale status 4 error protection	○		
	5	External scale status 5 error protection	○		
55	0	A-phase connection error protection	○		
	1	B-phase connection error protection	○		
	2	Z-phase connection error protection	○		
87	0	Compulsory alarm input protection		○	
95	0 to 4	Motor automatic recognition error protection			
Other number		Other error	○		

Note History...The error will be stored in the error history.

Can be cleared...To cancel the error, use the alarm clear input (A-CLR).

If the alarm clear input is not effective, turn off power, remove the cause of the error and then turn on power again.

Immediate stop...Instantaneous controlled stop upon occurrence of an error.

(Setting of "Pr.5.10 Sequence at alarm" is also required.)

- Caution**
- 1) Certain alarms are not included in the history. For detailed information on alarms e.g. alarm numbers, refer to P.6-2.
 - 2) When one of the errors which are listed in error history occurs, this error and history o shows the same error No.

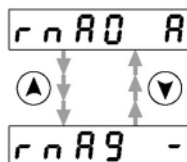
(8) Alarm Display

rn - -No alarm occurred

rn A0High priority alarm

Alarm number

- To display the alarm occurrence condition, press ▲ or ▼ button.



alarm No.	Alarm	Content	Latched time *1
A0	Overload protection	Load factor is 85% or more the protection level.	1 to 10s or ∞
A1	Over-regeneration alarm	Regenerative load factor is 85% or more the protection level.	10s or ∞
A2	Battery alarm	Battery voltage is 3.2 V or lower.	Fixed at ∞
A3	Fan alarm	Fan has stopped for 1 sec.	1 to 10s or ∞
A4	Encoder communication alarm	The number of successive encoder communication errors exceeds the specified value.	1 to 10s or ∞
A5	Encoder overheat alarm	The encoder detects overheat alarm.	1 to 10s or ∞
A6	Oscillation detection alarm	Oscillation or vibration is detected.	1 to 10s or ∞
A7	Lifetime detection alarm	Life expectancy of capacitor or fan is short.	Fixed at ∞
A8	External scale error alarm	The external scale detects the alarm.	1 to 10s or ∞
A9	External scale communication alarm	The number of successive external scale communication errors exceeds the specified value.	1 to 10s or ∞

*1 Alarms can be cleared by using the alarm clear. Because the all existing alarms are kept cleared while the alarm clear input (A-CLR) is kept ON, be sure to turn it OFF during normal operation. Either 1-10s or ∞ can be selected by using user parameter.

Exception: Battery alarm is fixed at ∞ because it is latched by the encoder.

Because the end of life alarm means that the life expectancy cannot be extended, the alarm is set at ∞.

(9) Display of Regenerative Load Factor, Over-load Factor and Inertia Ratio**• Regenerative Load Factor**

Display the ratio (%) against the alarm trigger level of regenerative protection.

This is valid when Pr0.16 (External regenerative resistor setup) is 0 or 1.

• Over-load Factor

Displays the ratio (%) against the rated load.

Refer to P.6-14, "Overload Protection Time Characteristics" of When in Trouble.

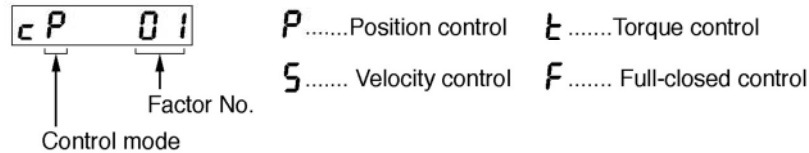
• Inertia Ratio

Displays the inertia ratio (%).

Value of Pr0.04 (Inertia Ratio) will be displayed as it is.

10) Display of the Factor of No-Motor Running

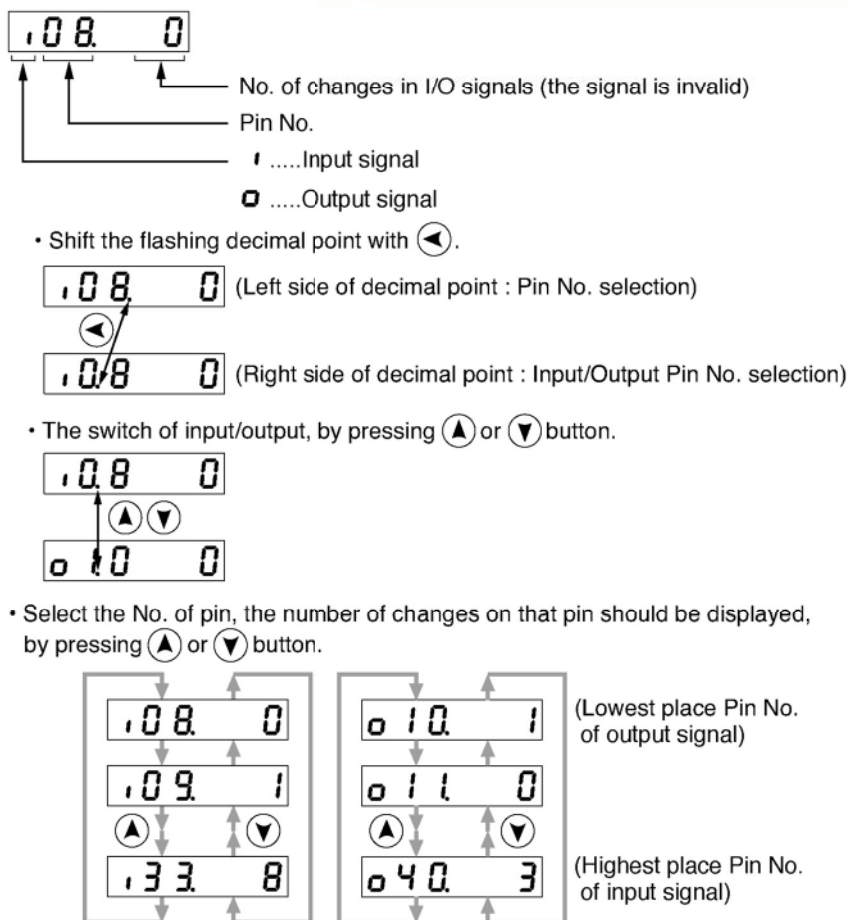
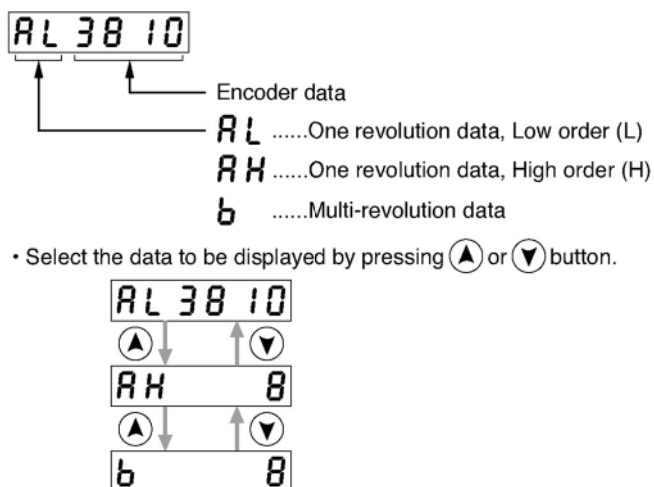
Displays the factor of no-motor running in number.



• Explanation of factor No.

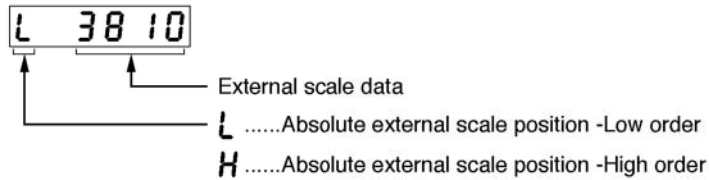
Factor No.	Factor	Related Control Mode				Content
		P	S	T	F	
flashing	Occurrence of error/alarm	○	○	○	○	An error is occurring, and an alarm is triggered.
00	No particular factor	○	○	○	○	No factor is detected for No-motor run. The motor runs in normal case.
01	Main power shutoff	○	○	○	○	The main power of the driver is not turned on.
02	No entry of SRV-ON input	○	○	○	○	The Servo-ON input (SRV-ON) is not connected to COM-.
03	Over-travel inhibition input is valid	○	○	○	○	While Pr5.04 is 0 (Run-inhibition input is valid), • Positive direction over-travel inhibition input (POT) is open and speed command is Positive direction. • Negative direction over-travel inhibition input (NOT) is open and speed command is Negative direction.
04	Torque limit setup is small	○	○	○	○	Either one of the valid torque limit setup value of Pr0.13 (1st) or Pr5.22 (2nd) is set to 5% or lower than the rating.
05	Analog torque limit input is valid.	○	○	○	○	While Pr5.21 is 0 (analog torque limit input accepted), • Positive direction analog torque limit input (P-ATL) is negative voltage and speed command is Positive direction. • Negative direction analog torque limit input (N-ATL) is positive voltage and speed command is Negative direction.
06	INH input is valid.	○	○	○	○	Pr5.18 is 0 (Command pulse inhibition input is valid.), and INH is open.
07	Command pulse input frequency is low.	○	○	○	○	The position command per each control cycle is 1 pulse or smaller due to, • No correct entry of command pulse • No correct connection to the input selected with Pr0.05. • No matching to input status selected with Pr0.06 pr Pr0.07.
08	CL input is valid.	○	○	○	○	While Pr5.17 is 0 (Deviation counter clear at level), the deviation counter clear input (CL) is connected to COM-.
09	ZEROSPD input is valid.	○	○	○	○	While Pr3.15 is 1 (Speed zero clamp is valid.), the speed zero clamp input (ZEROSPD) is open.
10	External speed command is small.	○	○	○	○	While the analog speed command is selected, the analog speed command is smaller than 0.06[V].
11	Internal speed command is 0.	○	○	○	○	While the internal speed command is selected, the internal speed command is set to lower than 30 [r/min]
12	Torque command is small.	○	○	○	○	The analog torque command input (SPR or P-ATL) is smaller than 5 [%] of the rating.
13	Speed limit is small.	○	○	○	○	• While Pr3.17 is 0 (speed is limited by 4th speed of internal speed), Pr3.07, (4th speed of speed setup) is set to lower than 30 [r/min]. • While Pr3.17 is 1 (speed is limited by SPR input), the analog speed limit input (SPR) is smaller than 0.06 [V].
14	Other factor	○	○	○	○	The motor runs at 20 [r/min] or lower even though the factors from 1 to 13 are cleared, (the command is small, the load is heavy, the motor lock or hitting, driver/motor fault etc.)

Note * Motor might run even though the other number than 0 is displayed.

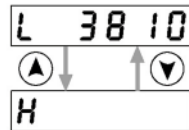
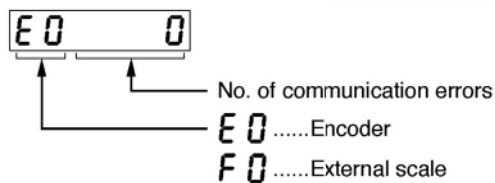
(11) Display of No. of changes in I/O signals**(12) Display of absolute encoder data**

(13) Display of absolute external scale position

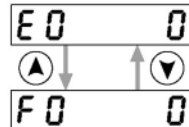
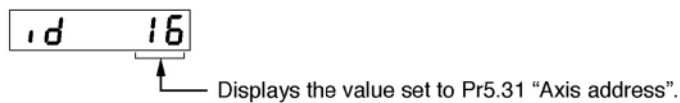
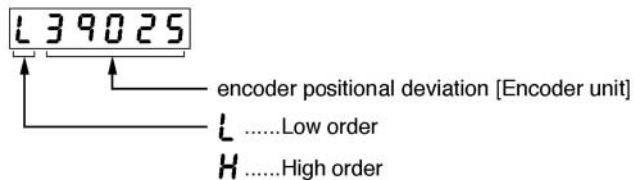
- Displays the absolute position of serial absolute scale.
- If a serial incremental scale, displays the scale position relative to the power on position which is defined as 0.



- Select encoder or external scale by pressing $\blacktriangle$ or $\blacktriangledown$ button.

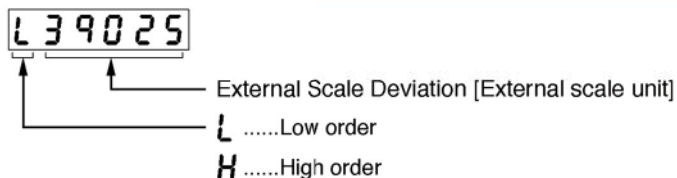
**(14) Display of No. of encoder/ external scale communication errors monitor**

- Select encoder or external scale by pressing $\blacktriangle$ or $\blacktriangledown$ button.

**(15) Display of communication axis address****(16) Display of encoder positional deviation [Encoder unit]**

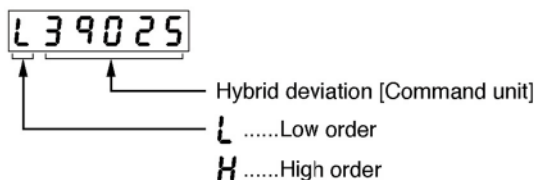
- To switch between Low order (L) and High order (H), press $\blacktriangleleft$.



(17) Display of External Scale Deviation [External Scale Unit]

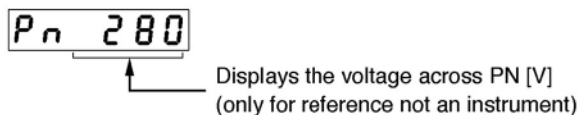
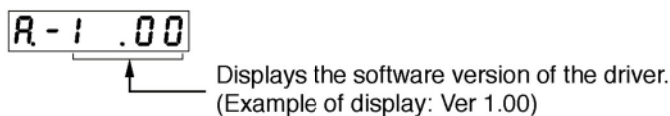
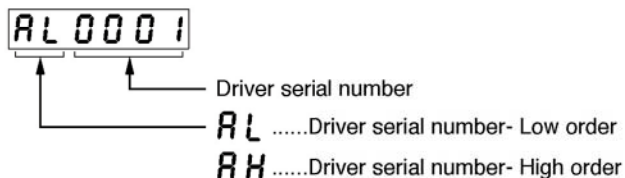
- To switch between Low order (L) and High order (H), press $\blacktriangleleft$.

L 39025 ↔ H 103

(18) Display of hybrid deviation [Command unit]

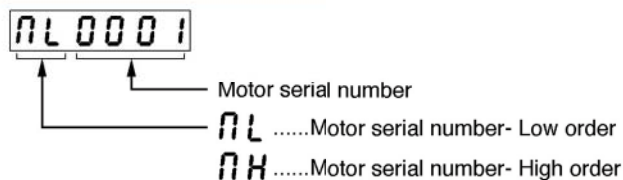
- To switch between Low order (L) and High order (H), press $\blacktriangleleft$.

L 39025 ↔ H 103

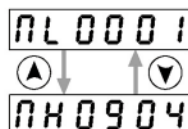
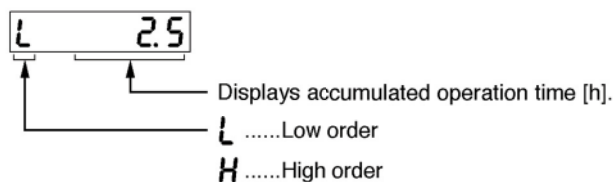
(19) Display of voltage across PN [V]**(20) Display of Software Version****(21) Display of driver serial number**

- To switch between Low order (L) and High order (H), press $\blacktriangleup$ or $\blacktriangledown$.
(Example of display: Serial number 09010001)

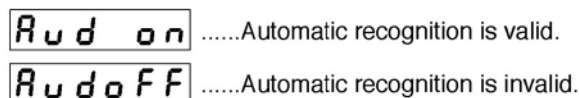
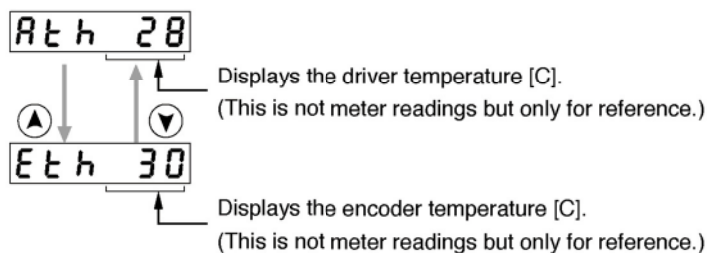
AL 0001
 $\blacktriangleup$ $\blacktriangledown$
 AH 0904

(22) Display of motor serial number

- To switch between Low order (L) and High order (H), press ▲ or ▼.
- (Example of display: Serial number 09040001)

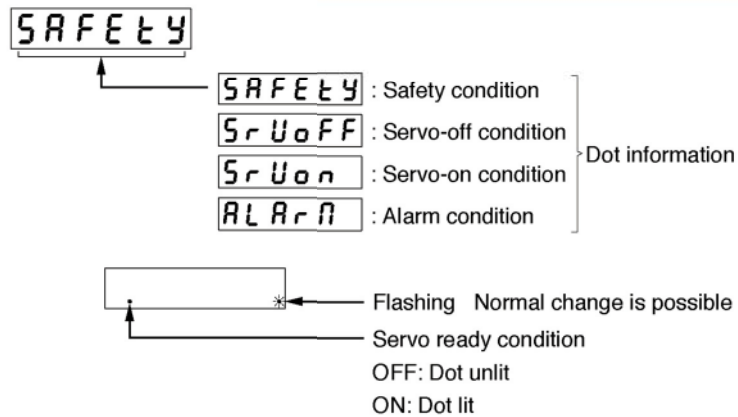
**(23) Display of accumulated operation time**

- To switch between Low order (L) and High order (H), press ◀.

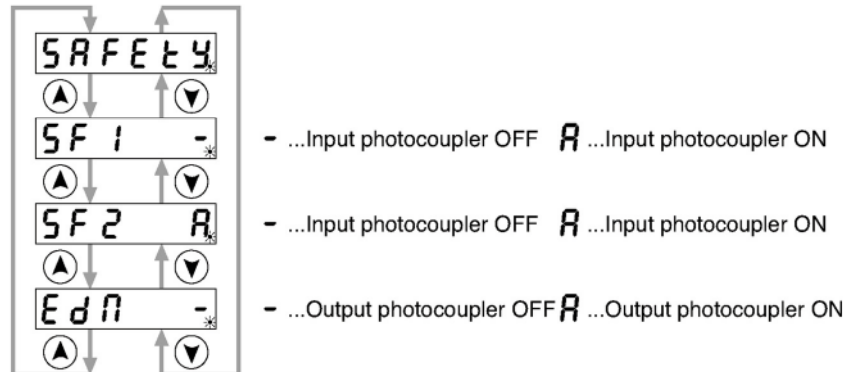
**(24) Automatic Motor Recognizing Function****(25) Display of temperature**

(*) Can be other values than 500 depending on the ZERO/SPAN adjustment.

(26) Display of safety condition monitor

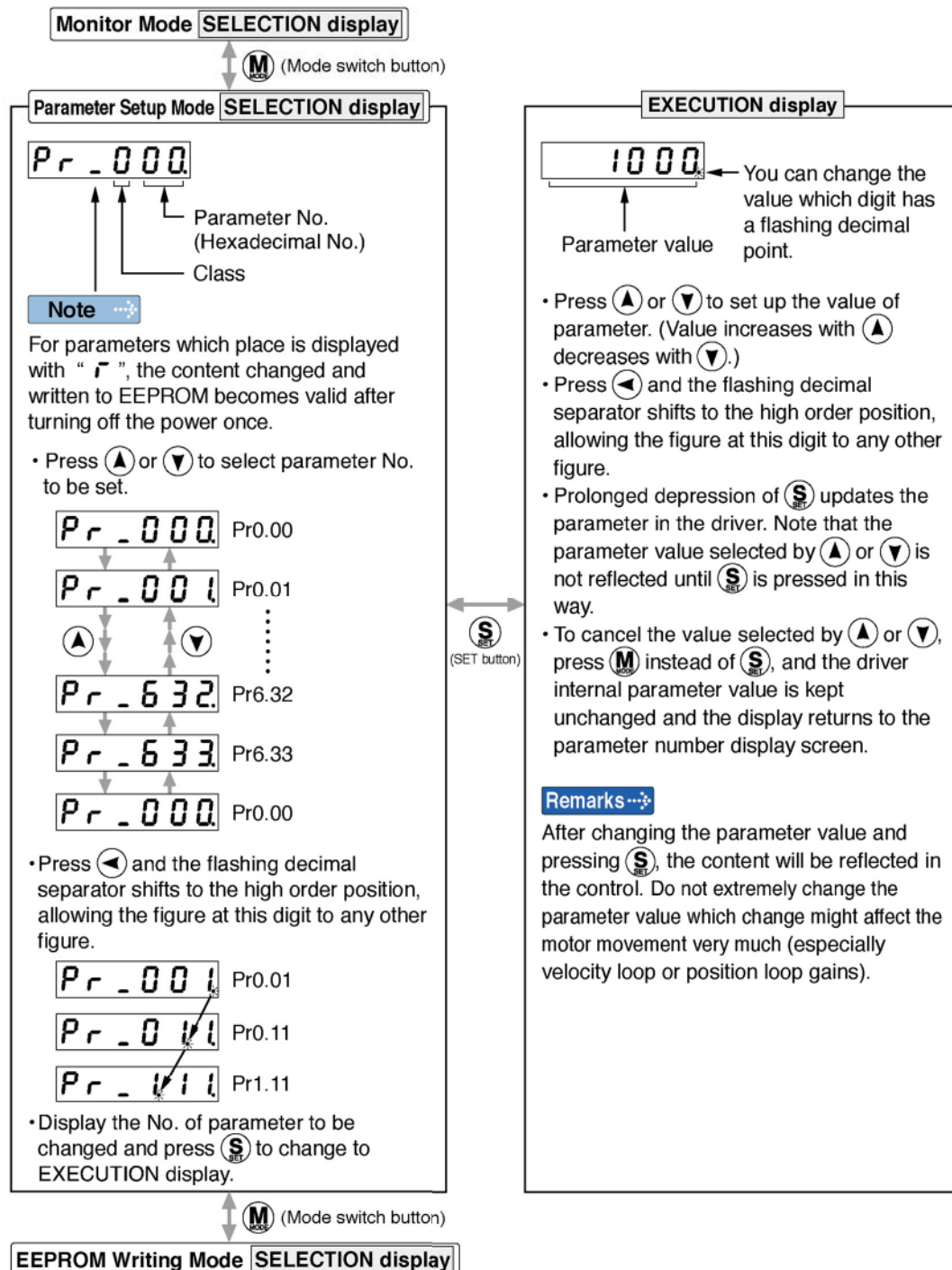


• Select desired monitor option by pressing ▲ or ▼ button.



How to Use the Front Panel

Parameter Setup Mode

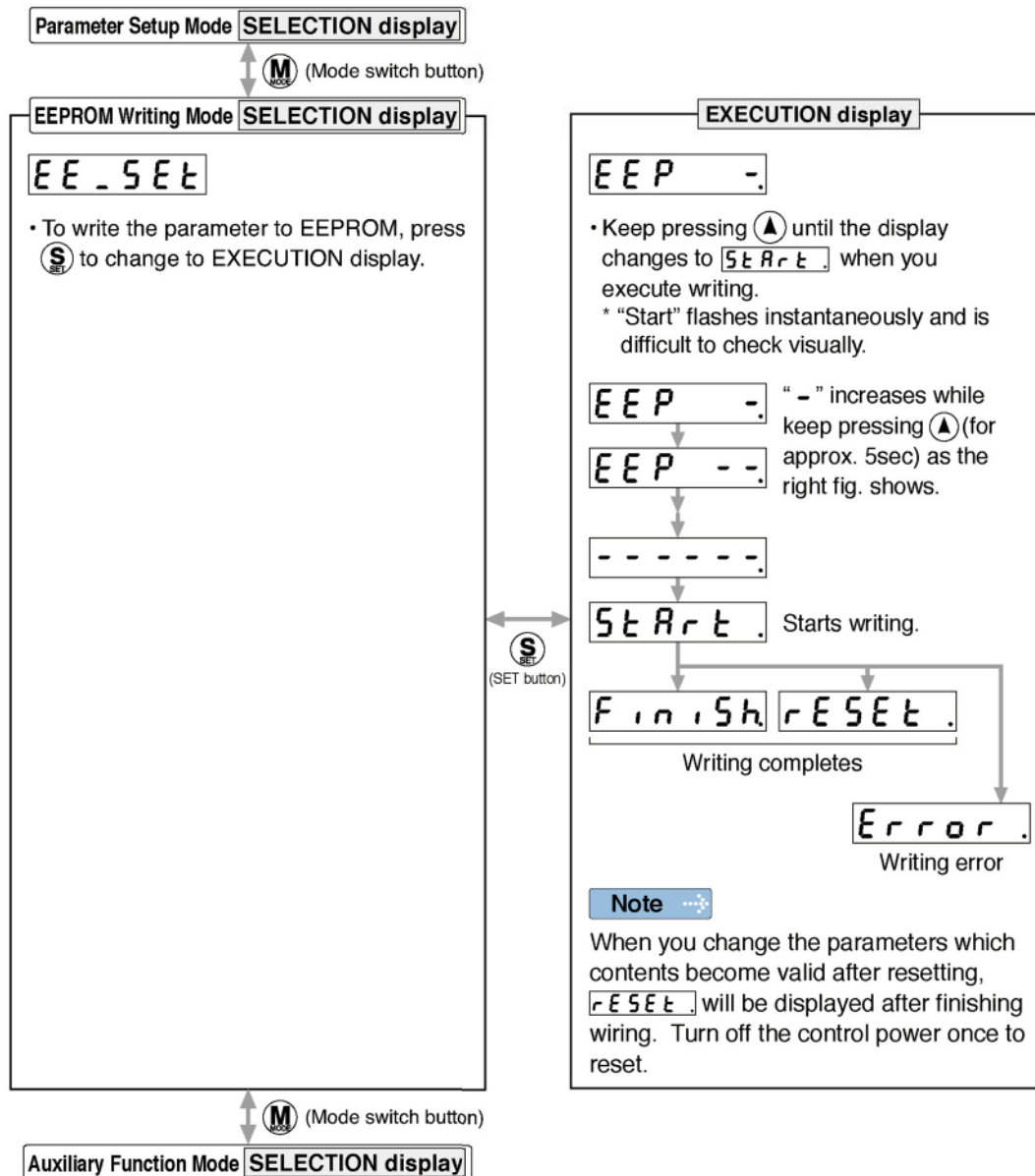


Note

- After setting up parameters, return to SELECT mode, referring to structure of each mode (P.2-88).
- Each parameter has a limit in number of places for upper-shifting.

How to Use the Front Panel

EPROM Writing Mode

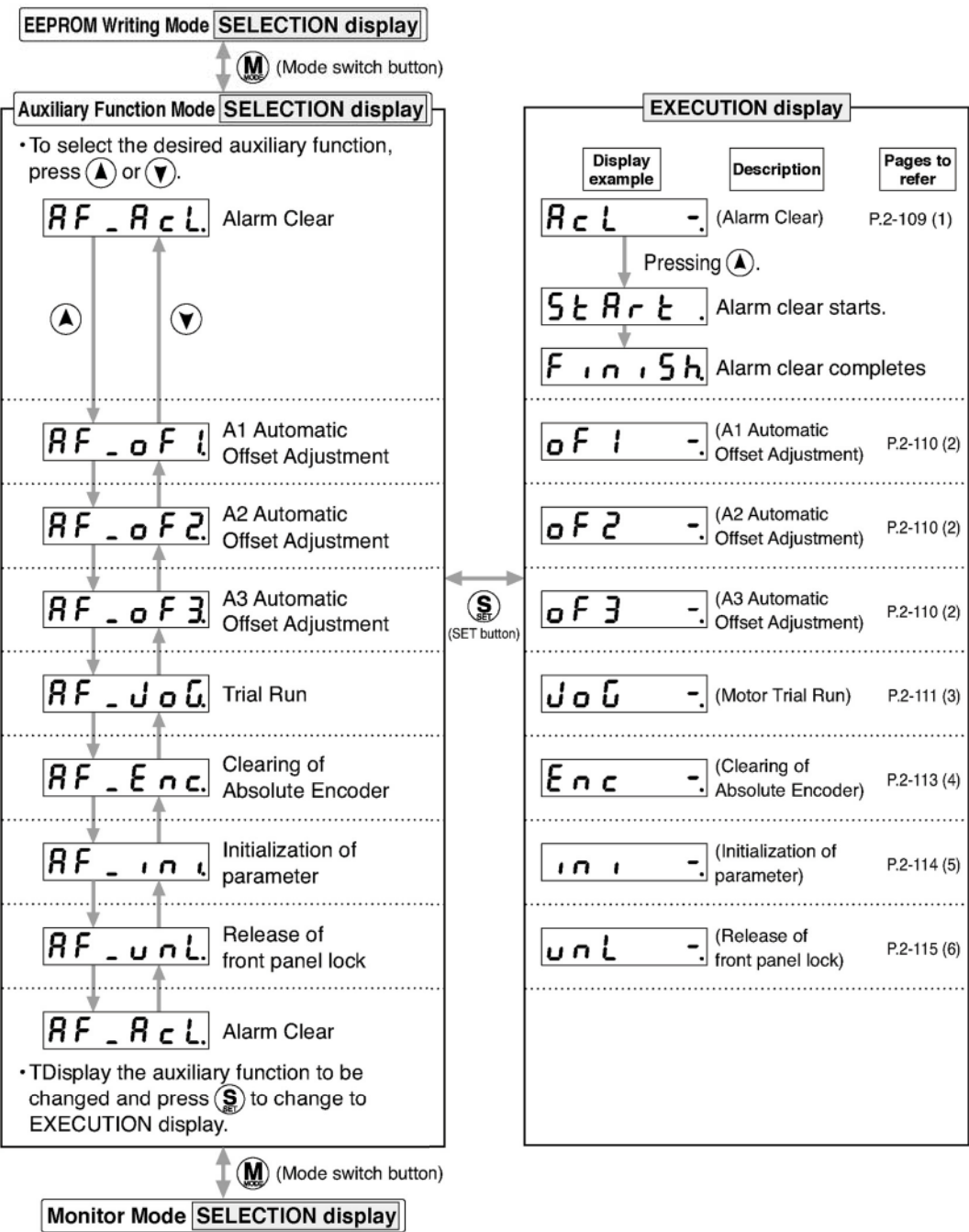


Caution

1. When writing error occurs, make writing again. If the writing error repeats many times, this might be a failure.
2. Don't turn off the power during EEPROM writing. Incorrect data might be written. If this happens, set up all of parameters again, and re-write after checking the data.
3. When the error defined by Err11.0 "Under voltage protection of control power supply" occurs, **Error** is displayed indicating that no writing is made to EEPROM.

How to Use the Front Panel

Auxilliary Function Mode (SELECTION display)



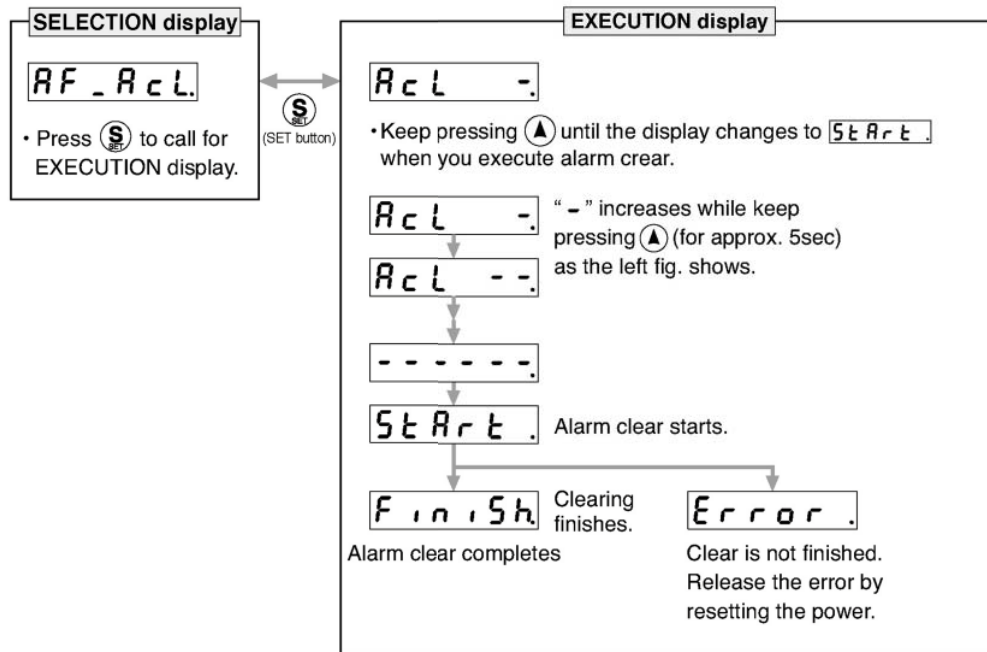
How to Use the Front Panel

Auxilliary Function Mode (EXECUTION display)

1) Alarm Clear Screen

This function releases the current alarm status.

Certain alarms will persist. If this is the case, refer to P.6-2 "When in Trouble - Protective Function".



Note

• After alarm cleaning, return to SELECTION display, referring to structure of each mode (P.2-88).

*) Can be other values than 500 depending on the ZERO/SPAN adjustment.

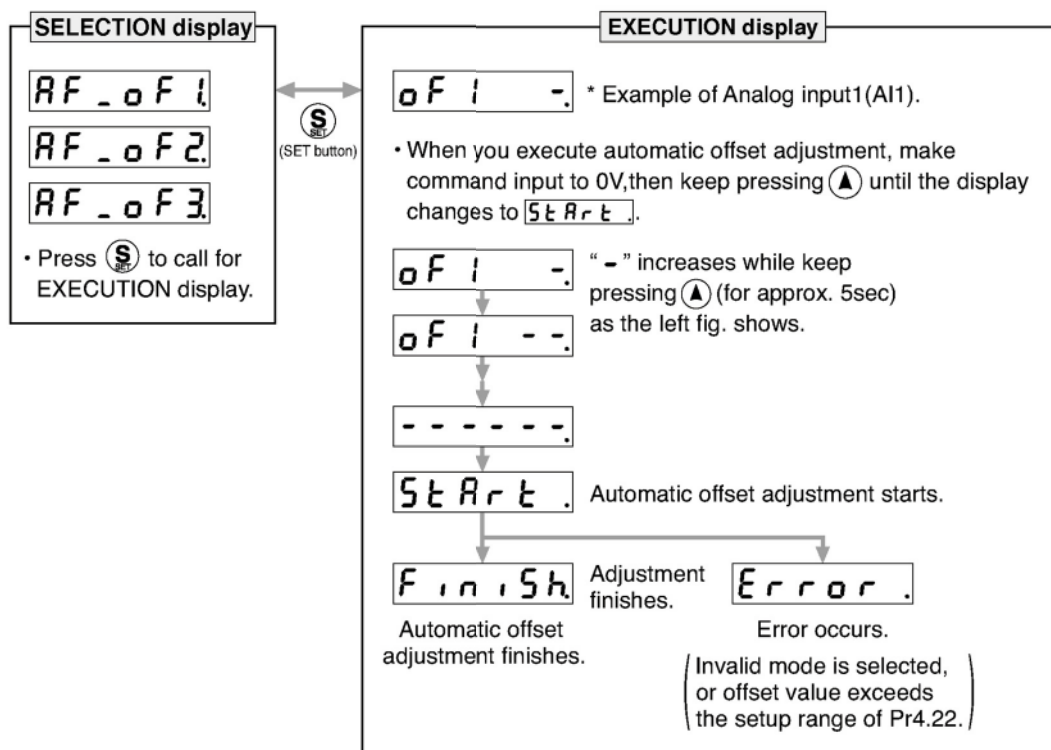
(2) Analog inputs 1 to 3 automatic offset adjustment

This function automatically adjusts offset setting of analog input.

Analog input 1 (AI1).....Pr4.22 (Analog input 1 (AI1) offset setup)

Analog Input 2 (AI2).....Pr4.25 (Analog input 2 (AI2) offset setup)

Analog input 3 (AI3).....Pr4.28 (Analog input 1 (AI3) offset setup)



Remarks

- You cannot write the data only by executing automatic offset adjustment. Execute a writing to EEPROM when you need to reflect the result afterward.

Note

- After completion of the automatic offset adjustment, return to SELECTION display by referring to P.2-88 “Structure of Each Mode”.

(3) Motor trial run

You can make a trial run (JOG run) without connecting the Connector, Connector X4 to the host controller such as PLC.

Remarks

- Separate the motor from the load, detach the Connector, Connector X4 before the trial run.
- Bring the user parameter setups (especially Pr0.04 and Pr1.01 to 1.04) to defaults, to avoid oscillation or other failure.

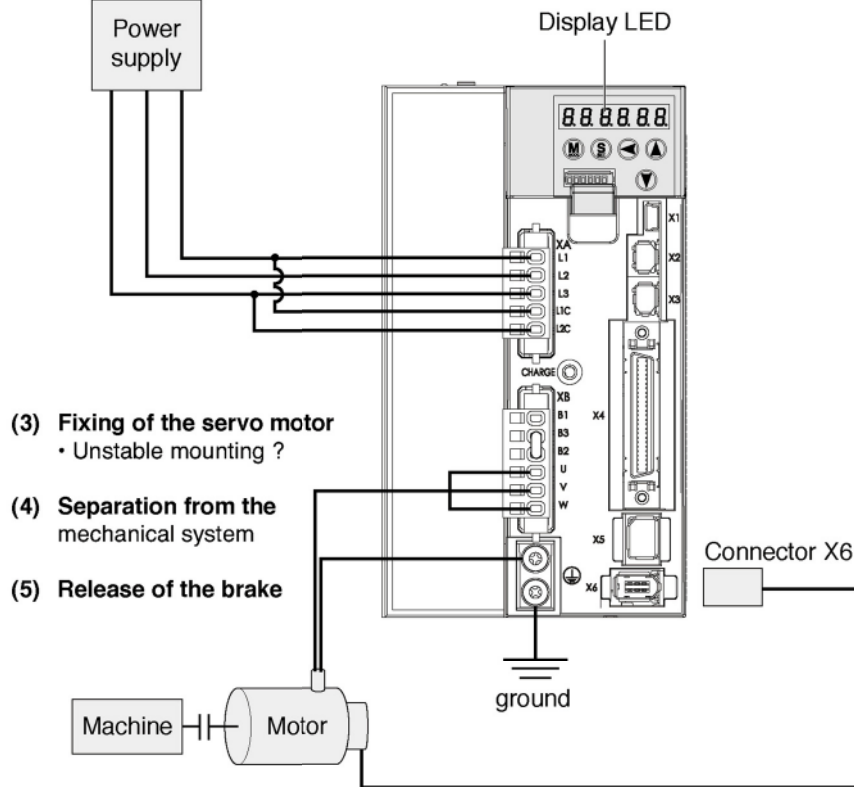
• Inspection Before Trial Run

(1) Inspection on wiring

- Miswiring ? (Especially power input and motor output)
- Short or grounded ?
- Loose connection ?

(2) Confirmation of power supply and voltage

- Rated voltage ?



(3) Fixing of the servo motor

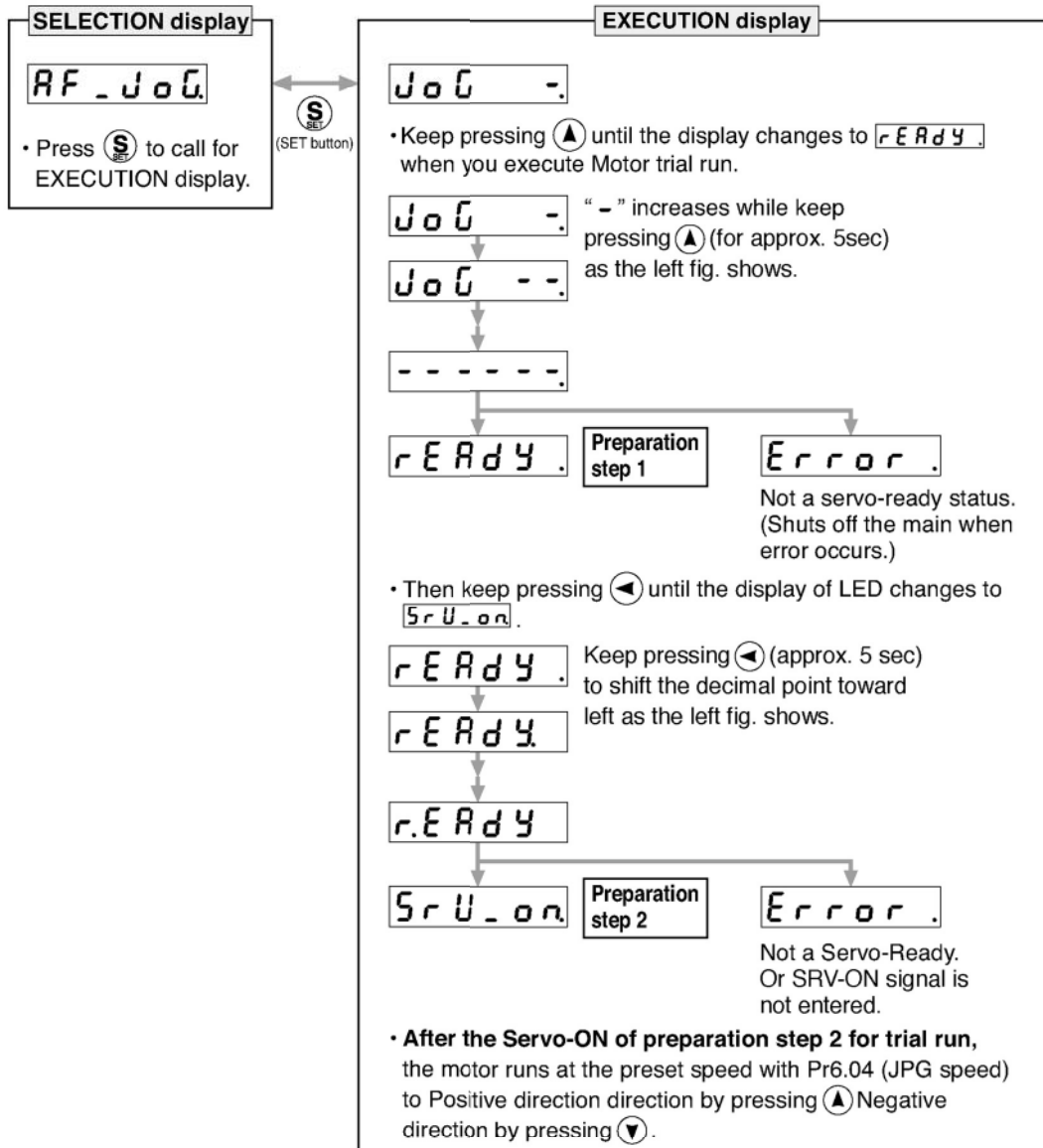
- Unstable mounting ?

(4) Separation from the mechanical system

(5) Release of the brake

(6) Turn to Servo-OFF after finishing the trial run by pressing .

• Procedure for Trial Run



Caution

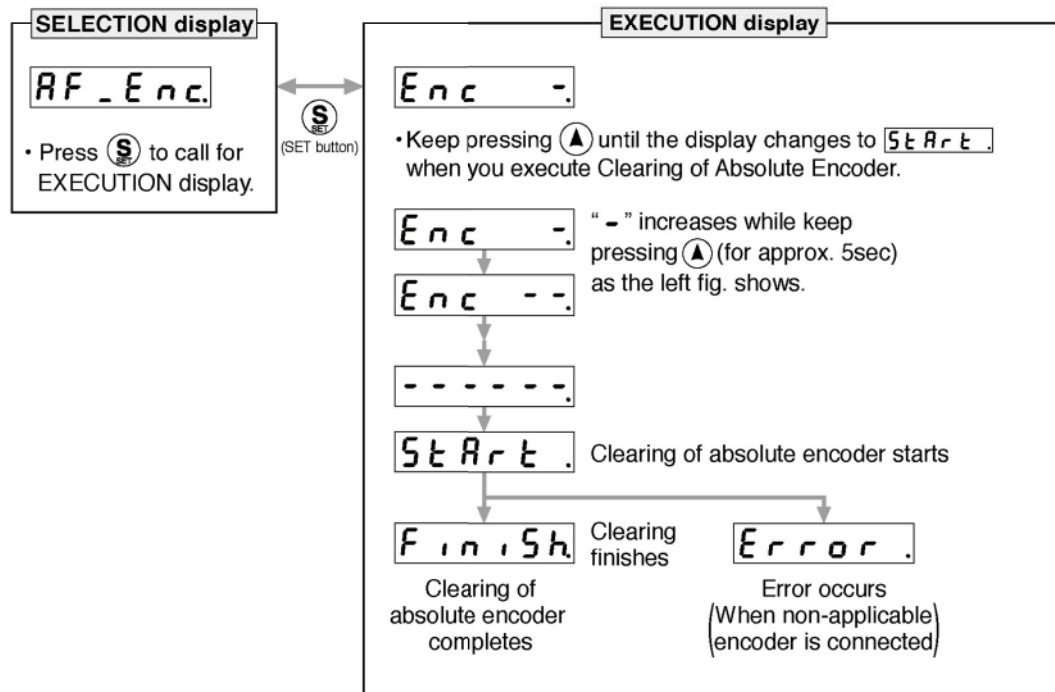
- Before starting the trial run, set the gain-related parameters to appropriate values to avoid problems such as oscillation. If the load is removed, be sure to set Pr0.04 “Inertia Ratio” to 0.
- During the trial run, use the velocity control mode. Various settings including parameters should assure safe and positive operation under appropriate velocity control.
- If SRV-ON becomes valid during trial run, the display changes to Error, which is normal run through external command.

Note

- After finished trial running, return to SELECTION display, referring to structure of each mode (P.2-88).

4) Clearing of Absolute Encoder

You can clear the multi-turn data of the absolute encoder.

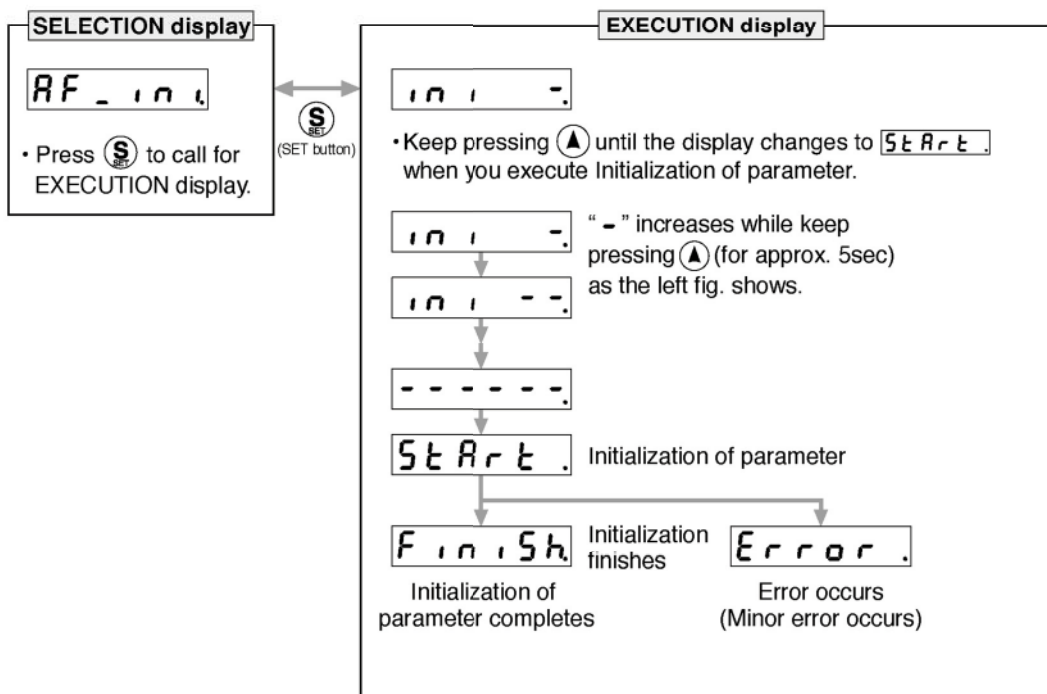


Note

• After clearing of absolute encoder finishes, return to SELECTION display, referring to structure of each mode (P.2-88).

(5) Initialization of parameter

Initialize the parameter.

**Caution**

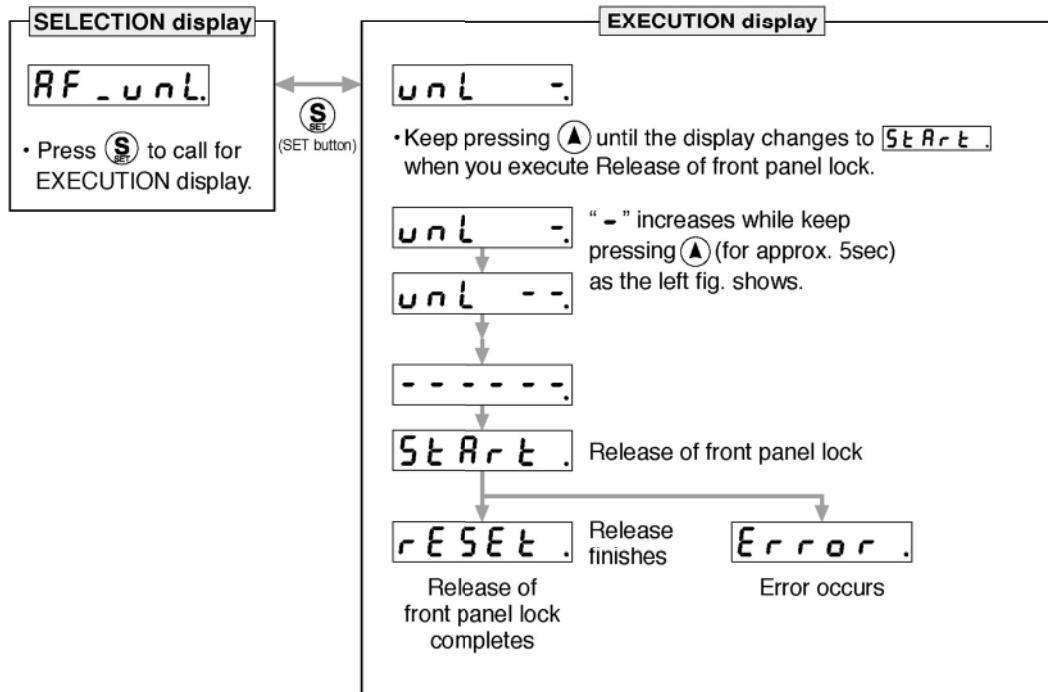
- Parameter cannot be initialized when one of the following error occurs: Err11.0 “Under voltage protection of control power supply”, EEPROM related errors (Err36.0, Err36.1, Err36.2, Err37.0, Err37.1 and Err37.2) - initialization will result in “Error” display.

Note

- After initialization of parameter finishes, return to SELECTION display, referring to structure of each mode (P.2-88).

(6) Release of front panel lock

Release the front panel lock setting.

**Note**

• After release of front panel lock finishes, return to SELECTION display, referring to structure of each mode (P.2-88).

AC Servo Driver Parameter

090617

ATLAS-204

• A parameter is designated as follows:

Pr0.00
Class ———— Parameter No.

• Definition of symbols under "Related mode"

P: position control, S: velocity control,
T: torque control, F: full closed control

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 0] Basic setting	00	Rotational direction setup	1	1	1	1	1	1	1
	01	Control mode setup	1	1	5	1	1	1	1
	02	Real-time auto-gain setup	0	0	0	0	0	0	0
	03	tuning selection of machine stiffness	12	11	11	12	12	12	12
	04	Inertia ratio	100	0	0	100	100	100	0
	05	Command pulse input selection	0	0	0	0	0	0	0
	06	rotational direction setup	0	0	0	0	0	0	0
	07	input mode setup	1	1	1	1	1	1	1
	08	Command pulse counts per one motor revolution	10000	10000	10000	10000	10000	10000	10000
	09	1st numerator of electronic gear	0	0	0	0	0	0	0
	10	Denominator of electronic gear	10000	10000	10000	10000	10000	10000	10000
	11	Output pulse counts per one motor revolution	1000	2500	1000	2500	2500	2500	250
	12	Reversal of pulse output logic	0	0	0	0	0	0	0
	13	1st torque limit	300	300	300	300	300	300	300
	14	Position deviation excess setup	488281	6400000	6400000	488281	488281	488281	6400000
	15	Absolute encoder setup	1	1	1	1	1	1	1
	16	External regenerative resistor setup	0	0	0	3	3	3	0
	17	Load factor of external regenerative resistor selection	0	0	0	0	0	0	0
[Class 1] Gain adjustment	00	1st gain of position loop	500	500	500	500	500	500	500
	01	gain of velocity loop	1000	1000	1000	1000	1000	1000	500
	02	time constant of velocity loop integration	500	10000	500	500	500	500	500
	03	filter of speed detection	4	4	4	4	4	4	4
	04	time constant of torque filter	50	100	50	50	50	50	100
	05	2nd gain of position loop	500	500	500	500	500	500	500
	06	gain of velocity loop	1000	1000	1000	1000	1000	1000	500
	07	time constant of velocity loop integration	500	5000	500	500	500	500	500
	08	filter of speed detection	4	4	4	4	4	4	4
	09	time constant of torque filter	50	50	50	50	50	50	50
	10	Velocity feed forward gain	0	0	0	0	0	0	0
	11	filter	0	0	0	0	0	0	0
	12	Torque feed forward gain	0	0	0	0	0	0	0
	13	filter	0	0	0	0	0	0	0
	14	2nd gain setup	0	1	0	0	0	0	0
	15	Position control mode	0	0	0	0	0	0	0
	16	switching delay time	0	0	0	0	0	0	0
	17	level	0	0	0	0	0	0	0
	18	hysteresis	0	0	0	0	0	0	0
	19	Position gain switching time	0	0	0	0	0	0	0
	20	Velocity control mode	0	5	0	0	0	0	0
	21	switching delay time	0	0	0	0	0	0	0
	22	level	0	5	0	0	0	0	0
	23	hysteresis	0	0	0	0	0	0	0
	24	Torque control mode	0	0	0	0	0	0	0
	25	switching delay time	0	0	0	0	0	0	0
	26	level	0	0	0	0	0	0	0
	27	hysteresis	0	0	0	0	0	0	0

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 2] Damping control	00	Adaptive filter mode setup	1	1	1	1	1	1	4
	01	1st notch	frequency	1000	5000	5000	5000	5000	5000
	02		width selection	2	2	2	2	2	2
	03		depth selection	0	0	0	0	0	0
	04	2nd notch	frequency	1000	1500	1500	1000	1000	5000
	05		width selection	2	2	2	2	2	2
	06		depth selection	0	0	0	0	0	0
	07	3rd notch	frequency	5000	562	562	5000	5000	2456
	08		width selection	2	2	2	2	2	2
	09		depth selection	0	0	0	0	0	0
	10	4th notch	frequency	5000	5000	5000	5000	5000	5000
	11		width selection	2	2	2	2	2	2
	12		depth selection	0	0	0	0	0	0
	13	Selection of damping filter switching	0	0	0	0	0	0	0
	14	1st damping	frequency	0	0	0	0	0	0
	15		filter setup	0	0	0	0	0	0
	16	2nd damping	frequency	0	0	0	0	0	0
	17		filter setup	0	0	0	0	0	0
	18	3rd damping	frequency	0	0	0	0	0	0
	19		filter setup	0	0	0	0	0	0
	20	4th damping	frequency	0	0	0	0	0	0
	21		filter setup	0	0	0	0	0	0
	22	Positional command	smoothing filter	2	2	2	2	2	2
	23		FIR filter	0	0	0	0	0	0
[Class 3] Velocity/Torque/ Full-closed control	00	Speed setup, Internal/External switching	0	0	0	0	0	0	0
	01	Speed command	rotational direction selection	0	0	0	0	0	0
	02		input gain	500(*)	200	400	500(*)	500(*)	300
	03		reversal input	1	0	0	1	0	1
	04	Speed setup	1st	0	0	0	0	0	0
	05		2nd	0	0	0	0	0	0
	06		3rd	0	0	0	0	0	0
	07		4th	0	0	1200	0	0	0
	08		5th	0	0	0	0	0	0
	09		6th	0	0	0	0	0	0
	10		7th	0	0	0	0	0	0
	11		8th	0	0	0	0	0	0
	12	Time setup	acceleration	0	0	0	0	0	0
	13		deceleration	0	0	0	0	0	0
	14	Sigmoid acceleration/deceleration time setup		0	0	0	0	0	0
	15	Speed zero-clamp	function selection	1	0	0	1	1	1
	16		level	30	30	30	30	30	30
	17	Torque command	selection	0	0	0	0	0	0
	18		direction selection	0	0	0	0	0	0
	19		input gain	30	30	30	30	30	30
	20		input reversal	0	0	0	0	0	0
	21	Speed limit value	1	0	0	0	0	0	0
	22		2	0	0	0	0	0	0
	23	External scale	selection	2	2	2	2	2	2
	24		numerator of division	0	0	0	0	0	0
	25		denominator of division	10000	10000	10000	10000	10000	10000
	26		reversal of direction	0	0	0	0	0	0
	27		Z phase disconnection detection disable	0	0	0	0	0	0
	28	Hybrid deviation	excess setup	122	1600	1600	122	122	1600
	29		clear setup	0	0	0	0	0	0

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 4] I/F monitor setting	00	Input selection	SI1 (Pin No.8)	8553090	8553090	8553090	8553090	8553090	8553090
	01		SI2 (Pin No.9)	8487297	8487297	8487297	8487297	8487297	8487297
	02		SI3 (Pin No.26)	9539850	9539850	9539850	9539850	9539850	9539850
	03		SI4 (Pin No.27)	394758	394758	394758	394758	394758	394758
	04		SI5 (Pin No.28)	4108	4108	4108	4108	4108	4108
	05		SI6 (Pin No.29)	197379	197379	197379	197379	197379	197379
	06		SI7 (Pin No.30)	3847	3847	3847	3847	3847	3847
	07		SI8 (Pin No.31)	263172	263172	263172	263172	263172	263172
	08		SI9 (Pin No.32)	328965	328965	328965	328965	328965	328965
	09		SI10 (Pin No.33)	3720	3720	3720	3720	3720	3720
	10	Output selection	SO1 (Pin No.10, 11 Line driver output)	197379	197379	197379	197379	197379	197379
	11		SO2 (Pin No.34, 35 Line driver output)	131586	131586	131586	131586	131586	131586
	12		SO3 (Pin No.36, 37 Line driver output)	65793	65793	65793	65793	65793	65793
	13		SO4 (Pin No.38, 39 Line driver output)	328964	328964	328964	328964	328964	328964
	14		SO5 (Pin No.12 Open collector output)	460551	460551	460551	460551	460551	460551
	15		SO6 (Pin No.40 Open collector output)	394758	394758	394758	394758	394758	394758
	16	Analog monitor 1	type	0	0	0	0	0	0
	17		output gain	500	500	500	500	500	500
	18	Analog monitor 2	type	4	4	4	4	4	4
	19		output gain	33	33	33	33	33	33
	20	Type of digital monitor		0	0	0	0	0	0
	21	Analog monitor output setup		0	0	0	0	0	0
	22	Analog input 1 (AI1)	offset setup	0	0	0	0	0	0
	23		filter	0	0	0	0	0	0
	24		overvoltage setup	0	0	0	0	0	0
	25	Analog input 2 (AI2)	offset setup	0	0	0	0	0	0
	26		filter	0	0	0	0	0	0
	27		overvoltage setup	0	0	0	0	0	0
	28	Analog input 3 (AI3)	offset setup	0	0	0	0	0	0
	29		filter	0	0	0	0	0	0
	30		overvoltage setup	0	0	0	0	0	0
	31	Positioning complete (In-position)	range	1	10	10	1	1	10
	32		output setup	0	0	0	0	0	0
	33	INP hold time		0	0	0	0	0	0
	34	Zero-speed		50	50	50	50	50	50
	35	Speed coincidence range		50	50	50	50	50	50
	36	At-speed (Speed arrival)		1000	1000	1000	1000	1000	1000
	37	Mechanical brake action	at stalling setup	0	0	0	0	0	0
	38		at running setup	0	0	0	0	0	0
	39	Mechanical brake action at running setup		30	30	30	30	30	30
	40	Selection of alarm output	1	0	0	0	0	0	0
	41		2	0	0	0	0	0	0
	42	2nd Positioning complete (In-position) range		10	10	10	10	10	10

Parameter No.		Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.									
[Class 5] Enhancing setting	00	2nd	numerator of electronic gear	0	0	0	0	0	0	0
	01	3rd	numerator of electronic gear	0	0	0	0	0	0	0
	02	4th	numerator of electronic gear	0	0	0	0	0	0	0
	03	Denominator of pulse output division		0	0	0	0	0	0	0
	04	Over-travel inhibit input setup		1	1	1	1	1	1	1
	05	Sequence at over-travel inhibit		0	0	0	0	0	0	0
	06	Sequence at Servo-Off		3	0	0	3	3	3	3
	07	Main power OFF	sequence	3	0	0	3	3	3	3
	08		LV trip selection	1	1	1	1	1	1	1
	09		detection time	70	70	70	70	70	70	70
	10	Sequence at alarm		3	0	0	3	3	3	3
	11	Torque setup for emergency stop		0	0	0	0	0	0	0
	12	Over-load level setup		0	0	0	0	0	0	0
	13	Over-speed level setup		0	0	0	0	0	0	0
	14	Motor working range setup		10	10	10	10	10	10	10
	15	I/F reading filter		0	0	0	0	0	0	0
	16	Alarm clear input setup		0	0	0	0	0	0	0
	17	Counter clear input mode		3	3	3	3	3	3	3
	18	Command pulse inhibit input	invalidation	1	1	1	1	1	1	1
	19		reading setup	4	4	4	4	4	4	4
	20	Position setup unit select		0	0	0	0	0	0	0
	21	Selection of torque limit		1	1	1	1	1	1	1
	22	2nd torque limit		300	300	300	300	300	300	300
	23	Torque limit switching setup	1	0	0	0	0	0	0	0
	24		2	0	0	0	0	0	0	0
	25	External input	positive direction torque limit	300	300	500	300	300	300	500
	26		negative direction torque limit	300	300	500	300	300	300	500
	27	Input gain of analog torque limit		30	30	30	30	30	30	30
	28	LED initial status		1	1	1	1	1	1	1
	29	RS232	baud rate setup	2	2	2	2	2	2	2
	30	RS485		2	2	2	2	2	2	2
	31	Axis address		1	1	1	1	1	1	1
	32	Command pulse input maximum setup		4000	4000	4000	4000	4000	4000	4000
	33	Pulse regenerative output limit setup		0	0	0	0	0	0	0
	34	For manufacturer's use		4	4	4	4	4	4	4
	35	Front panel lock setup		0	0	0	0	0	0	0

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 6] Special setting	00	Analog torque feed forward conversion gain	0	0	0	0	0	0	0
	02	Velocity deviation excess setup	0	0	0	0	0	0	0
	04	JOG trial run command speed	300	300	300	300	300	300	50
	05	Position 3rd gain	valid time	0	0	0	0	0	0
	06		scale factor	100	100	100	100	100	100
	07	Torque command additional value	0	0	0	0	0	0	0
	08	Positive direction	torque compensation	0	0	0	0	0	0
	09	Negative direction	value	0	0	0	0	0	0
	10	Function expansion setup	0	0	0	0	0	0	0
	11	Current response setup	100	100	100	100	100	100	100
	13	2nd Inertia ratio	250	250	250	250	250	250	250
	14	Emergency stop time at alarm	200	200	200	200	200	200	200
	15	2nd over-speed level setup	0	0	0	0	0	0	0
	17	Front panel parameter writing selection	0	0	0	0	0	0	0
	18	Power-up wait time	0	0	0	0	0	0	0
	19	Encoder Z phase setup	0	0	0	0	0	0	0
	20	Z-phase setup of external scale	0	0	0	0	0	0	0
	21	Serial absolute external scale Z phase setup	0	0	0	0	0	0	0
	22	A, B phase external scale pulse output method selection	0	0	0	0	0	0	0
	23	Disturbance torque compensating gain	0	0	0	0	0	0	0
	24	Disturbance observer filter	53	53	53	53	53	53	53
	27	Alarm latch time selection	1	5	5	1	1	1	1
	31	Real time auto tuning	estimation speed	1	1	1	1	1	1
	32		custom setup	0	0	0	0	0	0
	34	Hybrid vibration suppression	gain	0	0	0	0	0	0
	35		filter	10	10	10	10	10	10
	37	Oscillation detection level	0	0	0	0	0	0	0
	38	Alarm mask setup	0	0	4	0	0	0	0
	39	For manufacturer's use	0	0	0	0	0	0	0

ATLAS-234

- A parameter is designated as follows:

Pr0.00
Class ——— Parameter No.

- Definition of symbols under "Related mode"

P: position control, S: velocity control,
T: torque control, F: full closed control

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 0] Basic setting	00	Rotational direction setup	1	1	1	1	1	1	1
	01	Control mode setup	1	1	5	1	1	1	
	02	Real-time auto-gain tuning	0	0	0	0	0	0	
	03	selection of machine stiffness	12	11	11	12	12	12	12
	04	Inertia ratio	100	0	0	100	100	100	0
	05	Command pulse input selection	0	0	0	0	0	0	0
	06	rotational direction setup	0	0	0	0	0	0	0
	07	input mode setup	1	1	1	1	1	1	1
	08	Command pulse counts per one motor revolution	10000	10000	10000	10000	10000	10000	10000
	09	1st numerator of electronic gear	0	0	0	0	0	0	0
	10	Denominator of electronic gear	10000	10000	10000	10000	10000	10000	10000
	11	Output pulse counts per one motor revolution	1000	2500	1000	2500	2500	2500	250
	12	Reversal of pulse output logic	0	0	0	0	0	0	0
	13	1st torque limit	300	300	250	300	300	300	300
	14	Position deviation excess setup	488281	6400000	6400000	488281	488281	488281	6400000
	15	Absolute encoder setup	1	1	1	1	1	1	1
	16	External regenerative resistor setup	0	1	0	3	3	3	0
	17	Load factor of external regenerative resistor selection	0	0	0	0	0	0	0
[Class 1] Gain adjustment	00	1st gain of position loop	500	500	500	500	500	500	500
	01	gain of velocity loop	1000	1000	2000	1000	1000	1000	500
	02	time constant of velocity loop integration	500	10000	500	500	500	500	500
	03	filter of speed detection	4	4	4	4	4	4	4
	04	time constant of torque filter	50	100	50	50	50	50	100
	05	2nd gain of position loop	500	500	500	500	500	500	500
	06	gain of velocity loop	1000	1000	1000	1000	1000	1000	500
	07	time constant of velocity loop integration	500	5000	500	500	500	500	500
	08	filter of speed detection	4	4	4	4	4	4	4
	09	time constant of torque filter	50	50	50	50	50	50	50
	10	Velocity feed forward gain	0	0	0	0	0	0	0
	11	filter	0	0	0	0	0	0	0
	12	Torque feed forward gain	0	0	0	0	0	0	0
	13	filter	0	0	0	0	0	0	0
	14	2nd gain setup	0	1	0	0	0	0	0
	15	Position control mode	0	0	0	0	0	0	0
	16	switching delay time	0	0	0	0	0	0	0
	17	level	0	0	0	0	0	0	0
	18	hysteresis	0	0	0	0	0	0	0
	19	Position gain switching time	0	0	0	0	0	0	0
	20	Velocity control mode	0	5	0	0	0	0	0
	21	switching delay time	0	0	0	0	0	0	0
	22	level	0	5	0	0	0	0	0
	23	hysteresis	0	0	0	0	0	0	0
	24	Torque control mode	0	0	0	0	0	0	0
	25	switching delay time	0	0	0	0	0	0	0
	26	level	0	0	0	0	0	0	1
	27	hysteresis	0	0	0	0	0	0	0

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 2] Damping control	00	Adaptive filter mode setup	1	0	1	1	1	1	4
	01	1st notch	frequency	1000	220	580	5000	5000	5000
	02		width selection	2	2	2	2	2	2
	03		depth selection	0	30	0	0	0	0
	04	2nd notch	frequency	1000	5000	1600	1000	1000	5000
	05		width selection	2	2	2	2	2	2
	06		depth selection	0	0	0	0	0	0
	07	3rd notch	frequency	5000	5000	5000	5000	5000	2456
	08		width selection	2	2	2	2	2	2
	09		depth selection	0	0	0	0	0	0
	10	4th notch	frequency	5000	5000	5000	5000	5000	5000
	11		width selection	2	2	2	2	2	2
	12		depth selection	0	0	0	0	0	0
	13	Selection of damping filter switching	0	0	0	0	0	0	0
	14	1st damping	frequency	0	0	0	0	0	0
	15		filter setup	0	0	0	0	0	0
	16	2nd damping	frequency	0	0	0	0	0	0
	17		filter setup	0	0	0	0	0	0
	18	3rd damping	frequency	0	0	0	0	0	0
	19		filter setup	0	0	0	0	0	0
	20	4th damping	frequency	0	0	0	0	0	0
	21		filter setup	0	0	0	0	0	0
	22	Positional command	smoothing filter	2	2	2	2	2	2
	23		FIR filter	0	0	0	0	0	0
[Class 3] Velocity/Torque/ Full-closed control	00	Speed setup, Internal/External switching	0	0	0	0	0	0	0
	01	Speed command	rotational direction selection	0	0	0	0	0	0
	02		input gain	500(*)	250	400	500(*)	500(*)	300
	03		reversal input	1	0	0	1	0	1
	04	Speed setup	1st	0	0	0	0	0	0
	05		2nd	0	0	0	0	0	0
	06		3rd	0	0	0	0	0	0
	07		4th	0	0	0	0	0	0
	08		5th	0	0	0	0	0	0
	09		6th	0	0	0	0	0	0
	10		7th	0	0	0	0	0	0
	11		8th	0	0	0	0	0	0
	12	Time setup	acceleration	0	0	0	0	0	0
	13		deceleration	0	0	0	0	0	0
	14	Sigmoid acceleration/deceleration time setup		0	0	0	0	0	0
	15	Speed zero-clamp	function selection	1	0	0	1	1	1
	16		level	30	30	30	30	30	30
	17	Torque command	selection	0	0	1	0	0	0
	18		direction selection	0	0	0	0	0	0
	19		input gain	30	30	30	30	30	30
	20		input reversal	0	0	1	0	0	0
	21	Speed limit value	1	0	0	0	0	0	0
	22		2	0	0	0	0	0	0
	23	External scale	selection	2	2	2	2	2	2
	24		numerator of division	0	0	0	0	0	0
	25		denominator of division	10000	10000	10000	10000	10000	10000
	26		reversal of direction	0	0	0	0	0	0
	27		Z phase disconnection detection disable	0	0	0	0	0	0
	28	Hybrid deviation	excess setup	122	1600	1600	122	122	1600
	29		clear setup	0	0	0	0	0	0

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 4] I/F monitor setting	00	Input selection	SI1 (Pin No.8)	8553090	8553090	8553090	8553090	8553090	8553090
	01		SI2 (Pin No.9)	8487297	8487297	8487297	8487297	8487297	8487297
	02		SI3 (Pin No.26)	9539850	9539850	9539850	9539850	9539850	9539850
	03		SI4 (Pin No.27)	394758	394758	394758	394758	394758	394758
	04		SI5 (Pin No.28)	4108	4108	4108	4108	4108	4108
	05		SI6 (Pin No.29)	197379	197379	197379	197379	197379	197379
	06		SI7 (Pin No.30)	3847	3847	3847	3847	3847	3847
	07		SI8 (Pin No.31)	263172	263172	263172	263172	263172	263172
	08		SI9 (Pin No.32)	328965	328965	328965	328965	328965	328965
	09		SI10 (Pin No.33)	3720	3720	3720	3720	3720	3720
	10	Output selection	SO1 (Pin No.10, 11 Line driver output)	197379	197379	197379	197379	197379	197379
	11		SO2 (Pin No.34, 35 Line driver output)	131586	131586	131586	131586	131586	131586
	12		SO3 (Pin No.36, 37 Line driver output)	65793	65793	65793	65793	65793	65793
	13		SO4 (Pin No.38, 39 Line driver output)	328964	328964	328964	328964	328964	328964
	14		SO5 (Pin No.12 Open collector output)	460551	460551	460551	460551	460551	460551
	15		SO6 (Pin No.40 Open collector output)	394758	394758	394758	394758	394758	394758
	16	Analog monitor 1	type	0	0	0	0	0	0
	17		output gain	500	500	500	500	500	500
	18	Analog monitor 2	type	4	4	4	4	4	4
	19		output gain	33	33	33	33	33	33
	20	Type of digital monitor		0	0	0	0	0	0
	21	Analog monitor output setup		0	0	0	0	0	0
	22	Analog input 1 (AI1)	offset setup	0	0	0	0	0	0
	23		filter	0	0	0	0	0	0
	24		overvoltage setup	0	0	0	0	0	0
	25	Analog input 2 (AI2)	offset setup	0	0	0	0	0	0
	26		filter	0	0	0	0	0	0
	27		overvoltage setup	0	0	0	0	0	0
	28	Analog input 3 (AI3)	offset setup	0	0	0	0	0	0
	29		filter	0	0	0	0	0	0
	30		overvoltage setup	0	0	0	0	0	0
	31	Positioning complete (In-position)	range	1	10	10	1	1	10
	32		output setup	0	0	0	0	0	0
	33	INP hold time		0	0	0	0	0	0
	34	Zero-speed		50	50	50	50	50	50
	35	Speed coincidence range		50	50	50	50	50	50
	36	At-speed (Speed arrival)		1000	1000	1000	1000	1000	1000
	37	Mechanical brake action	at stalling setup	0	0	0	0	0	0
	38		at running setup	0	0	0	0	0	0
	39	Mechanical brake action at running setup		30	30	30	30	30	30
	40	Selection of alarm output	1	0	0	0	0	0	0
	41		2	0	0	0	0	0	0
	42	2nd Positioning complete (In-position) range		10	10	10	10	10	10

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 5] Enhancing setting	00	2nd	0	0	0	0	0	0	0
	01	3rd	0	0	0	0	0	0	0
	02	4th	0	0	0	0	0	0	0
	03	Denominator of pulse output division	0	0	0	0	0	0	0
	04	Over-travel inhibit input setup	1	1	1	1	1	1	1
	05	Sequence at over-travel inhibit	0	0	0	0	0	0	0
	06	Sequence at Servo-Off	3	0	0	3	3	3	3
	07	Main power OFF	3	0	0	3	3	3	3
	08	sequence	1	1	1	1	1	1	1
	09	LV trip selection	70	70	70	70	70	70	70
	10	detection time	3	0	0	3	3	3	3
	11	Sequence at alarm	0	0	0	0	0	0	0
	12	Torque setup for emergency stop	0	0	0	0	0	0	0
	13	Over-load level setup	0	0	0	0	0	0	0
	14	Over-speed level setup	0	0	0	0	0	0	0
	15	Motor working range setup	10	10	10	10	10	10	10
	16	I/F reading filter	0	0	0	0	0	0	0
	17	Alarm clear input setup	0	0	0	0	0	0	0
	18	Counter clear input mode	3	3	3	3	3	3	3
	19	Command pulse	1	1	1	1	1	1	1
	20	invalidation	4	4	4	4	4	4	4
	21	reading setup	0	0	0	0	0	0	0
	22	Position setup unit select	1	1	1	1	1	1	1
	23	Selection of torque limit	300	300	250	300	300	300	300
	24	2nd torque limit	0	0	0	0	0	0	0
	25	Torque limit	0	0	0	0	0	0	0
	26	switching setup	300	300	250	300	300	300	500
	27	External input	300	300	250	300	300	300	500
	28	positive direction torque limit	30	30	30	30	30	30	30
	29	negative direction torque limit	1	1	1	1	1	1	1
	30	Input gain of analog torque limit	2	2	2	2	2	2	2
	31	LED initial status	2	2	2	2	2	2	2
	32	RS232	1	1	1	1	1	1	1
	33	RS485	4000	4000	4000	4000	4000	4000	4000
	34	Axis address	0	0	0	0	0	0	0
	35	Command pulse input maximum setup	4	4	4	4	4	4	4
		Pulse regenerative output limit setup	0	0	0	0	0	0	0
		For manufacturer's use	0	0	0	0	0	0	0
		Front panel lock setup							

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 6] Special setting	00	Analog torque feed forward conversion gain	0	0	0	0	0	0	0
	02	Velocity deviation excess setup	0	0	0	0	0	0	0
	04	JOG trial run command speed	300	300	300	300	300	300	50
	05	Position 3rd gain	0	0	0	0	0	0	0
	06	valid time	100	100	100	100	100	100	100
	07	scale factor	0	0	0	0	0	0	0
	08	Torque command additional value	0	0	0	0	0	0	0
	09	Positive direction	0	0	0	0	0	0	0
	10	Negative direction	0	0	0	0	0	0	0
	11	torque compensation value	0	0	0	0	0	0	0
	13	Function expansion setup	0	0	0	0	0	0	0
	14	Current response setup	100	100	100	100	100	100	100
	15	2nd Inertia ratio	250	250	250	250	250	250	250
	17	Emergency stop time at alarm	200	200	200	200	200	200	200
	18	2nd over-speed level setup	0	0	0	0	0	0	0
	19	Front panel parameter writing selection	0	0	0	0	0	0	0
	20	Power-up wait time	0	0	0	0	0	0	0
	21	Encoder Z phase setup	0	0	0	0	0	0	0
	22	Z-phase setup of external scale	0	0	0	0	0	0	0
	23	Serial absolute external scale Z phase setup	0	0	0	0	0	0	0
	24	A, B phase external scale pulse output method selection	0	0	0	0	0	0	0
	27	Disturbance torque compensating gain	0	0	0	0	0	0	0
	31	Disturbance observer filter	53	53	53	53	53	53	53
	32	Alarm latch time selection	1	5	5	1	1	1	1
	34	Real time auto tuning	1	1	1	1	1	1	
	35	estimation speed	0	0	0	0	0	0	0
	37	custom setup	0	0	0	0	0	0	0
	38	Hybrid vibration suppression	0	0	0	0	0	0	0
	39	gain	10	10	10	10	10	10	10
		filter	0	0	0	0	0	0	0
		Oscillation detection level	0	0	0	0	0	0	0
		Alarm mask setup	0	0	4	0	0	0	0
		For manufacturer's use	0	0	0	0	0	0	0